

Role of Interfaces in Solid-State Batteries

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Solid-state batteries (SSBs) are considered as one of the most promising candidates for the next-generation energy-storage technology, because they simultaneously exhibit high safety, high energy density, and wide operating temperature range. The replacement of liquid electrolytes with solid electrolytes produces numerous solid–solid interfaces within the SSBs. A thorough understanding on the roles of these interfaces is indispensable for the rational performance optimization. In this review, the interface issues in the SSBs, including internal buried interfaces within solid electrolytes and composite electrodes, and planar interfaces between electrodes and solid electrolyte separators or current collectors are discussed. The challenges and future directions on the investigation and optimization of these solid–solid interfaces for the production of the SSBs are also assessed.

1. Introduction

The present commercial rechargeable lithium (Li)-ion batteries (LIBs) with liquid electrolytes have promoted the development of portable electronics, but the fast-growing markets of electric vehicles and grid-scale energy storage require even higher energy density and significantly improved safety.^[1–6] Due to the flammability and limited electrochemical stability of the organic liquid electrolyte, current LIBs are difficult to meet the two aforementioned requirements.^[5,6] Nevertheless, replacing the liquid electrolyte with solid electrolytes (SEs) to construct the solid-state batteries (SSBs) is a highly promising solution.^[7–13] Most SEs are nonflammable and more durable. As a result, they can circumvent the safety issues associated with organic liquid electrolytes.^[12–23] Besides, the excellent electrochemical stability of SEs may allow for the use of electrodes with much higher specific capacities. A typical example is the Li metal anode, which shows the highest theoretical capacity (3860 mA h g^{−1}, ten times higher than that of the currently used graphite anode) and the lowest electrochemical potential (−3.040 V vs standard hydrogen electrode).^[24,25] Because of these advantages, SSBs

have rapidly become the focal point of both academia and industry.

In order to realize high-performance SSBs, various SEs have been developed, including polymer SEs, inorganic SEs, and composite solid electrolytes (CSEs).^[12–23,26–28] Polymer SEs feature high flexibility, facile manufacturability, excellent conformability, and low density, but they usually exhibit much lower room-temperature ionic conductivities (e.g., $\approx 10^{-7}$ to 10^{-5} S cm^{−1} for poly(ethylene oxide) (PEO) SEs) than conventional liquid electrolytes ($\approx 10^{-2}$ S cm^{−1}).^[26–28] Inorganic SEs (such as oxides, sulfides, and halides) possess high ionic conductivities of $\approx 10^{-4}$ to 10^{-2} S cm^{−1} at room temperature (e.g., $\approx 10^{-2}$ S cm^{−1} for Li_{9.54}Si_{1.74}P_{1.44}S_{11.7}Cl_{0.3} and Li₁₀GeP₂S₁₂) and Li-ion transference numbers close to unity.^[15–19,29,30] However, they often suffer from nondeformability and/or poor stability in ambient air. In comparison, CSEs combine the advantages of inorganic and polymer SEs, so they have recently attracted significant research interest.^[12,21–23] No matter which kind of SE is used, its existence in the SSBs inevitably introduces various solid–solid interfaces that are absent in commercial LIBs. As a result, different challenges arise.

In conventional LIBs, liquid electrolytes can easily permeate into the whole interpenetrating porous structure composed of the composite electrodes and porous separator, forming effective, percolative ion-transport pathways among cathode and anode active material particles (Figure 1a). With the liquid electrolyte and porous separator replaced by SEs, these liquid–solid interfaces become solid–solid interfaces (Figure 1b). Besides, the SEs themselves may exhibit different types of “internal interfaces” too, including grain boundaries (for polycrystalline SEs) and interphase boundaries (for CSEs). The solid–solid interfaces whose properties are quite different from those of liquid–solid ones frequently act as the bottleneck for the overall ion transport (Figure 1c,d).^[31–33] For example, grain boundaries in inorganic SEs may impede the ion transport and reduce the total ionic conductivity by orders of magnitude.^[15,16] The poor solid–solid contact or undesired reactions between the SEs and the electrode active materials can result in large interfacial resistance, leading to low Coulombic efficiency and considerable performance degradation of the entire battery.^[12–23,31,32] Evidently, the interfaces in SSBs involve complex physical, (electro)chemical, and electrochemical–mechanical processes. An in-depth understanding on their roles and microscopic mechanisms is critical to the construction of decent solid–solid interfaces.

In this review, we analyze the interface issues in SSBs, including both the internal buried interfaces within SEs (such as grain boundaries and interphase boundaries) and the planar

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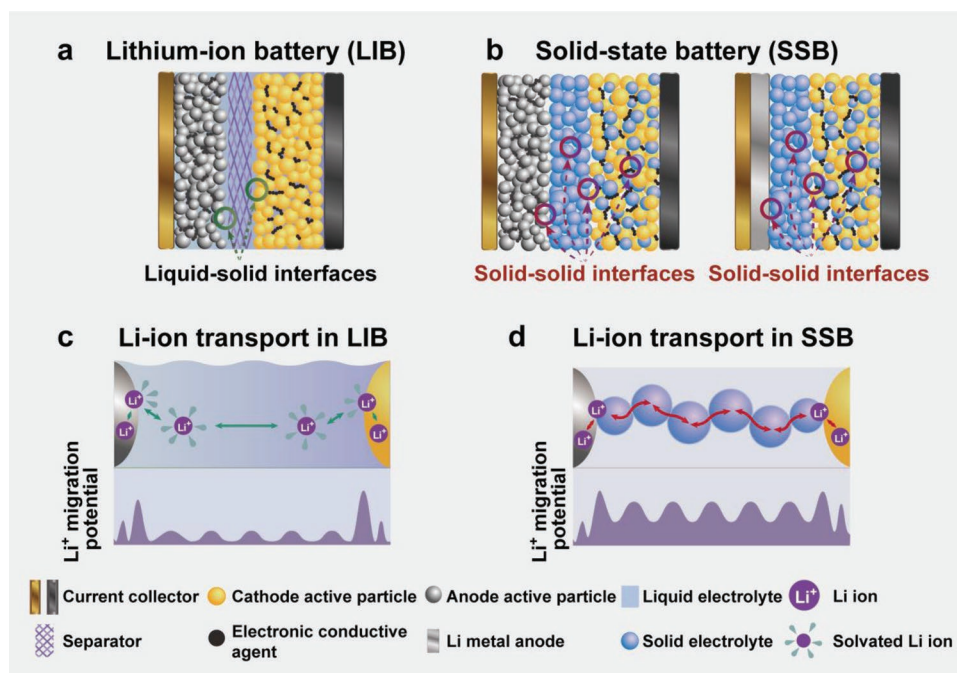


Figure 1. Change from liquid–solid interfaces to solid–solid interfaces. a) An LIB cell consisting of multiphase composite electrode layers and a porous polymer separator layer with liquid electrolyte soaked through the whole cell. b) An SSB cell consisting either of multiphase composite electrode layers and an SE layer (left), or of a composite cathode layer, an SE layer, and a Li-metal anode (right). Schematic illustration for the Li-ion transport in c) LIBs and d) SSBs. The Li⁺ migration potential increases at each interface.

ones between the electrodes and the SE separator or the current collectors. The influence of these interfaces, the mechanisms of interfacial interactions, the existing challenges, and the corresponding strategies for improvement are discussed in detail. At the end, future perspectives on the design, characterization, and optimization of the interfaces in SSBs are outlined.

2. Interfaces within Solid Electrolytes

In comparison with liquid electrolytes, the SEs are mostly inhomogeneous with various internal interfaces, such as the grain boundaries in polycrystalline SEs, the interfaces between inorganic fillers and the polymer matrix in CSEs, and the interfaces between the crystalline regions and the amorphous ones in glass-ceramic SEs or even in semicrystalline polymer electrolytes. All these interfaces may affect the ion-transport behaviors in SEs, and, in some cases, the influence could be decisive.

2.1. Grain Boundaries in Inorganic Solid Electrolytes

As a common feature for polycrystalline materials, grain boundaries (GBs) are present in most inorganic SEs.^[15,16] A proper understanding on their roles in ion transport is crucial to both mechanism investigation and performance optimization. The contribution of the grains and GBs to the ion transport in inorganic SEs is usually characterized by using electrochemical impedance spectroscopy. For example, the semicircles in the high and medium frequency regions in the Nyquist plot of an oxide electrolyte represent the impedance components

originating from the grains and GBs, respectively, as shown in Figure 2a. Nevertheless, for many SEs, the GBs do not induce a separate, easily distinguishable semicircle in the Nyquist plot.^[34,35] In this situation, the GB resistance needs to be determined by deconvoluting each component using the equivalent circuits constructed according to the observed capacitances.^[15,36] for a pellet with 1 cm diameter and 1 mm thickness, the capacitance ranges for bulk, GB, and interfacial impedances are usually considered to lie in $\approx 10^{-11}$, $\approx 10^{-10}$ to 10^{-7} , and $\approx 10^{-7}$ to 10^{-5} F, respectively, during the assignment of circuit elements.^[36,37] Following this principle, the GB resistance can be determined for SEs without an easily distinguishable GB semicircle in the Nyquist plot, such as $\text{Li}_6\text{PS}_5\text{Cl}$ and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$.^[34,35] However, it should be noted that the aforementioned capacitance ranges and the construction of equivalent circuits are largely based on empirical hypotheses.^[15] Consequently, when the GB semicircle is obscure, the corresponding resistances reported by different research groups often vary considerably.^[15]

Depending on the type of materials, the influence of GBs could vary significantly. For the sulfide and halide SEs with high ionic conductivity, the GBs do not create much hindrance to Li-ion migration.^[17,19] Thanks to the high deformability of these materials, a cold pressing under 100–300 MPa can directly produce rather intimate contact (and thus efficient ion transport) among the crystallites.^[38,39] The total ionic conductivity of such cold-pressed pellets may easily reach 10^{-4} to 10^{-2} S cm⁻¹ at room temperature (RT), and the associated GB resistance is usually not high enough to contribute a separate semicircle in the Nyquist plot (Figure 2a). When the pressure used in cold pressing is low (e.g., 50 MPa), the high porosity of sulfide SEs could generate larger GB impedance, leading to a

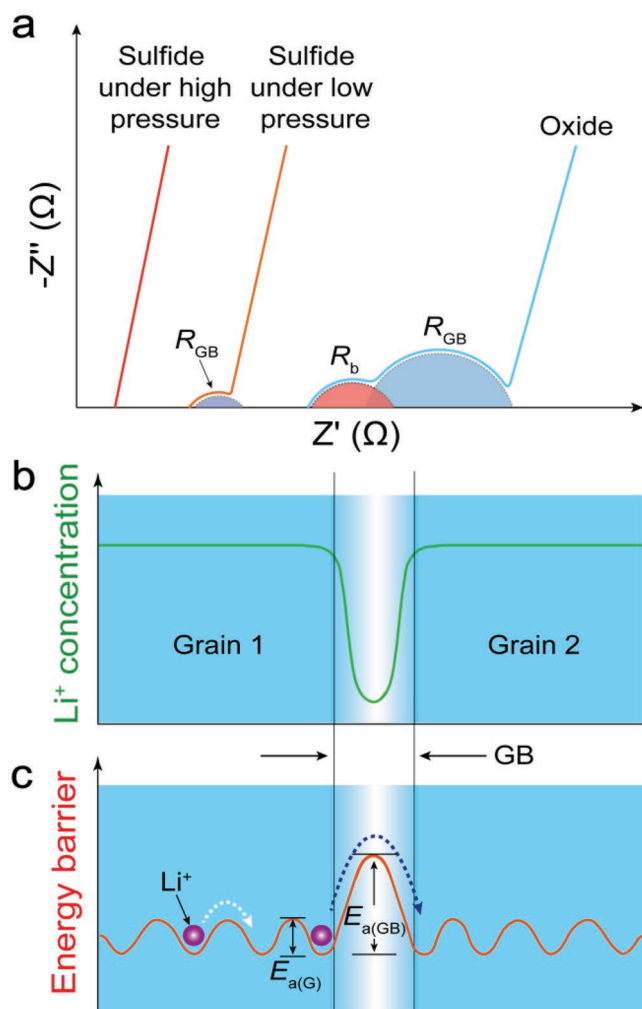


Figure 2. a) Typical Nyquist plots of sulfide (or halide) SEs under different pressures and that of oxide SEs. R_b and R_{GB} represent the grain bulk resistance and GB resistance, respectively. Schematic illustration of the variation of b) Li^+ concentration and c) energy barrier across the GB. $E_{a(G)}$ and $E_{a(GB)}$ represent the activation energy in the grains and GBs, respectively.

semicircle in the Nyquist plot (Figure 2a).^[40–44] However, there are few reports on the effect of GBs in sulfide and halide SEs so far.^[45,46]

For oxide SEs, the situation is different. This category of materials mainly includes NASICON-type $\text{Li}_{1+x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$, perovskite-type $(\text{Li}_{3-x}\text{La}_{2/3-x})\text{TiO}_3$ (LLTO), and garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO).^[15,16,18,47] Unlike sulfides and halides, most (if not all) of the mainstream oxide SEs are rather brittle, and the particles cannot plastically deform under pressure. Consequently, the cold pressing alone may only lead to poor intercrystallite contact, and the resulting pellet would not exhibit meaningful ionic conductivity either. In order to address this issue, the cold-pressed SE needs to be sintered at high temperatures (typically above 1100 °C), where the densification process can make the crystallites intimately contact each other through the formation of GBs.^[15,16,18,47] However, such GBs are frequently associated with large resistance and severely impede the overall ion transport. For example, although the bulk conductivity of

the perovskite SEs like $\text{Li}_{0.33}\text{La}_{0.56}\text{TiO}_3$ is as high as $10^{-3} \text{ S cm}^{-1}$, the overly large GB resistance lowers the total ionic conductivity to the $10^{-5} \text{ S cm}^{-1}$ level.^[47,48] Similarly, the NASICON-type SEs such as $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$ may also reach a rather high bulk ionic conductivity of $10^{-3} \text{ S cm}^{-1}$, but the total ionic conductivity is one order of magnitude lower due to the large GB resistance.^[47] For oxide SEs, the Li-ion concentration at the GB is usually lower than that of the grain bulk (Figure 2b), leading to the increase of the energy barrier for Li-ion migration (Figure 2c).^[48] Although the ion transport in most oxide SEs is significantly impeded by GBs, exceptions do exist. For the LLZO-type SEs such as $\text{Li}_{6.25}\text{Al}_{0.25}\text{La}_3\text{Zr}_2\text{O}_{12}$, the GB and bulk conductivities are both around $10^{-4} \text{ S cm}^{-1}$, so the former would not significantly compromise the total ionic conductivity.^[49] Regardless, such oxide SEs are rare.^[15,16,18,47] In addition to the GBs directly connecting two adjacent grains, it is worth noting that impurities and secondary phases would affect the migration of Li-ions at the GBs of LLZO too.^[50,51] For example, certain side reactions between LLZO and moisture in the environment would initiate at the GBs and generate poorly ionically conductive impurities that disrupt the migration of Li-ions as well as the integrity of the electrolyte.^[50]

In order to improve the GB ion transport, different approaches have been developed. For example, LiF was introduced into $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$ to inhibit the generation of the impurity phase Li_2CO_3 and stabilize the Li-Al-O glass phase at the GBs.^[52] SiO_2 assisted with the formation of amorphous GBs in the LLTO electrolyte, which eliminated the anisotropy of the grain shell and reduced the Li^+ diffusion potential barrier.^[53] Boron (B), a common glass network forming agent, was adopted to form glassified GBs in the $\text{LiTi}_2(\text{PO}_4)_3$ glass-ceramic electrolyte, which substantially reduced the interfacial resistance at the GBs with little effect on bulk resistance, and thus the conductivity of the B-doped glass-ceramic electrolyte was much larger than that of the undoped one ($1.03 \times 10^{-3} \text{ S cm}^{-1}$ vs $1.12 \times 10^{-4} \text{ S cm}^{-1}$).^[54] To tune the properties of the space charge layer at GBs, doping is an effective way, the main idea of which is enhancing the carrier/dopant concentration and reducing the space charge potential. For example, Nb-doping was applied to the LLTO system and the GB conductivity of LLTO was improved by a factor of 2–3.^[55]

The origin of the different GB behaviors may provide effective guiding principles for the conductivity optimization. Unfortunately, at present this issue is not fully understood, due to the complex structures (including defects) of GBs and the absence of effective characterization methods. As highly localized features, the GBs are usually no thicker than 1–2 unit cells.^[48,56] In order to probe their structures, the experimental investigation must employ characterization techniques with high spatial resolution, such as the aberration-corrected scanning transmission electron microscopy (C_s -corrected STEM). Nevertheless, most SEs are highly sensitive to the intense electron-beam irradiation needed to conduct atomic-resolution observation.^[57,58] This challenge greatly limits the number of SEs that can be studied. At present, only the perovskite-structured oxide SEs have been systematically studied by C_s -corrected STEM.^[48,59] Other systems are too electron-beam-sensitive to be examined at the atomic scale; instead, they can at most be characterized by high-resolution transmission electron microscopy (HRTEM),

which merely discloses information regarding the lattice symmetry and is not indicative of the atomic structure.^[56,60,61] Despite their rareness in literature, the reported experimental studies on the atomic configurations of GBs still provide important insights. According to the high-angle annular dark-field STEM observation, the LLTO SE undergoes local structural reconstruction at most of its GBs.^[48] The reconstructed GB no longer exhibits the perovskite structure, but is essentially a Ti-O layer that barely allows for the existence or transport of Li-ions. In comparison, such local reconstruction does not occur in the garnet-structured SEs.^[62,63] The HRTEM images clearly demonstrate that their grains almost join directly with each other, whereas the electron energy-loss spectra do not indicate any fundamental difference between the atomic configuration of GBs and that of the bulk either.^[63] Consistent with this observation, the garnet-structured SEs are not plagued by the large GB resistance.^[49] Therefore, the local reconstruction seems a key phenomenon to the sluggish GB ion transport. The reason why this behavior occurs to the perovskite system but not the garnet system remains an open question. A potential cause might lie in the symmetry. The garnet-structured SEs exhibit the cubic symmetry, while the perovskite-structured ones crystallize in tetragonal or orthorhombic symmetries (the “cubic” structure that was reported for the perovskite SEs before has already been demonstrated to be a short-range-ordered tetragonal one).^[47,49,64] It is possible that the higher symmetry makes the garnet structure more capable of adapting to the orientation difference between adjacent grains, and thus largely eliminates the need for significant local structural reconstruction. Since presently such in-depth characterization cannot be performed to other SEs, it is in fact unclear whether the reported mechanism is applicable to most systems, and the potential existence of other mechanisms that may also contribute to the large GB resistance cannot be ruled out either. In order to reach a definitive answer, similar investigation must be conducted to as many SEs as possible, including the sulfide and halide systems showing highly efficient GB ion transport. Considering that the electron-beam irradiation damage of the nonoxide systems is even more severe than that of the oxides, the development of low-dose electron microscopy techniques that allow for non-damage atomic-resolution observation at RT would be crucial to the study of this subject.^[57]

2.2. Interfaces within Composite Solid Electrolytes

In order to realize high energy densities, a thin SE separator layer (e.g., 10–25 μm in thickness) is required. Reaching such a small thickness with pure inorganic SEs would be rather difficult for bulk SSBs. In comparison, the CSEs may easily achieve this goal. Regardless, the rational optimization of the CSEs still demands a thorough understanding on their internal buried interfaces, especially the ones between the inorganic fillers and the polymer matrix (Figure 3a).

The incorporation of inorganic particles in polymer creates interactions between the particles and the polymer chains located in the vicinity of the particles (Figure 3a).^[12,21–23] According to Lewis’ model, each particle is surrounded by the Stern layer and the diffuse Gouy–Chapman layer or the

space charge layer.^[65,66] The Lewis acid–base-type interactions between the filler surface and polymer chains kinetically inhibit the crystallization of polymer chains and enhance chain segmental motion, which effectively facilitates the Li-ion transport.^[21,22,67] Besides, the Lewis acid–base interactions with Li salts promote the dissociation of Li salts and increase the concentration of free Li^+ near the interface region.^[21,22,67–69] Therefore, the percolative interfacial layers (the inset in Figure 3b) due to the Lewis acid–base interactions establish fast ion-transport pathways with both a high concentration of mobile Li-ions n and a large ion mobility μ ($\mu \propto \exp(-E_a/(k_B T))$, where E_a is the conduction activation energy, k_B is the Boltzmann constant, and T is the temperature).^[12] This is why the introduction of a small number of inorganic fillers can often decrease E_a and enhance the ionic conductivity σ ($\sigma = q \times n \times \mu$, where q is the Li-ion charge), considerably facilitating the ion transport (Figure 3b,c).

The inorganic fillers can be either inert particles that do not conduct Li-ions by themselves (such as Al_2O_3 and SiO_2) or active fillers that are inorganic SEs.^[12] The incorporation of both kinds of fillers may induce the Lewis acid–base interactions as discussed above. The difference between the inert fillers and the active ones is that the inorganic SE particles as active fillers can also provide additional fast ion transport pathways via themselves, leading to higher ionic conductivity. However, if the surfaces of the SE particles are not fresh and contain poorly conductive species, the ionic conductivity of the resulting CSE would be compromised. For example, LiOH and Li_2CO_3 can easily form on the LLZO particle surfaces through the reaction with moisture and carbon dioxide in air.^[8,52] If such particles are used as the filler, no expected ionic conductivity can be achieved. On the other hand, the tightly bound Stern layer at the inorganic SE particle/matrix interface may shield the SE particles from ion conduction. In this case, inorganic SE particles act like common inert fillers. Generally speaking, the creation of highly conductive interface layers between the filler and the polymer matrix is crucial for the successful fabrication of the CSEs.

Many experiments and simulations have been devoted to revealing the ion-transport mechanism in the polymer-matrix CSEs, but this issue is still not well understood; in particular, details about the complex interactions between the inorganic fillers and the polymer matrix remain elusive. Although the absence of such knowledge prevents the establishment of systematic guiding principles for materials optimization, some effective strategies are still developed, as discussed below.

Various inorganic fillers have been used in polymer electrolytes to obtain desired interface layers. The fillers with a high specific surface area, such as nanoparticles^[70] and nanoporous materials (e.g., mesoporous silica,^[71,72] porous Al_2O_3 ,^[73] and metal–organic frameworks^[74]), are more likely to produce more interfacial layers that may serve as percolative pathways for ion transport at a small loading of fillers.^[12] For example, adding Al_2O_3 or TiO_2 nanoparticles in the PEO/ LiClO_4 electrolytes significantly improved the ionic conductivity by over two orders of magnitude.^[70] However, the agglomeration or aggregation of the fillers would destroy the percolation pathways along the interfacial layers. In order to take full advantage of the interfacial layers, it is necessary to form a continuous percolated network with the maximized interfacial layer volume fraction but prevent the agglomeration effect in the meantime.

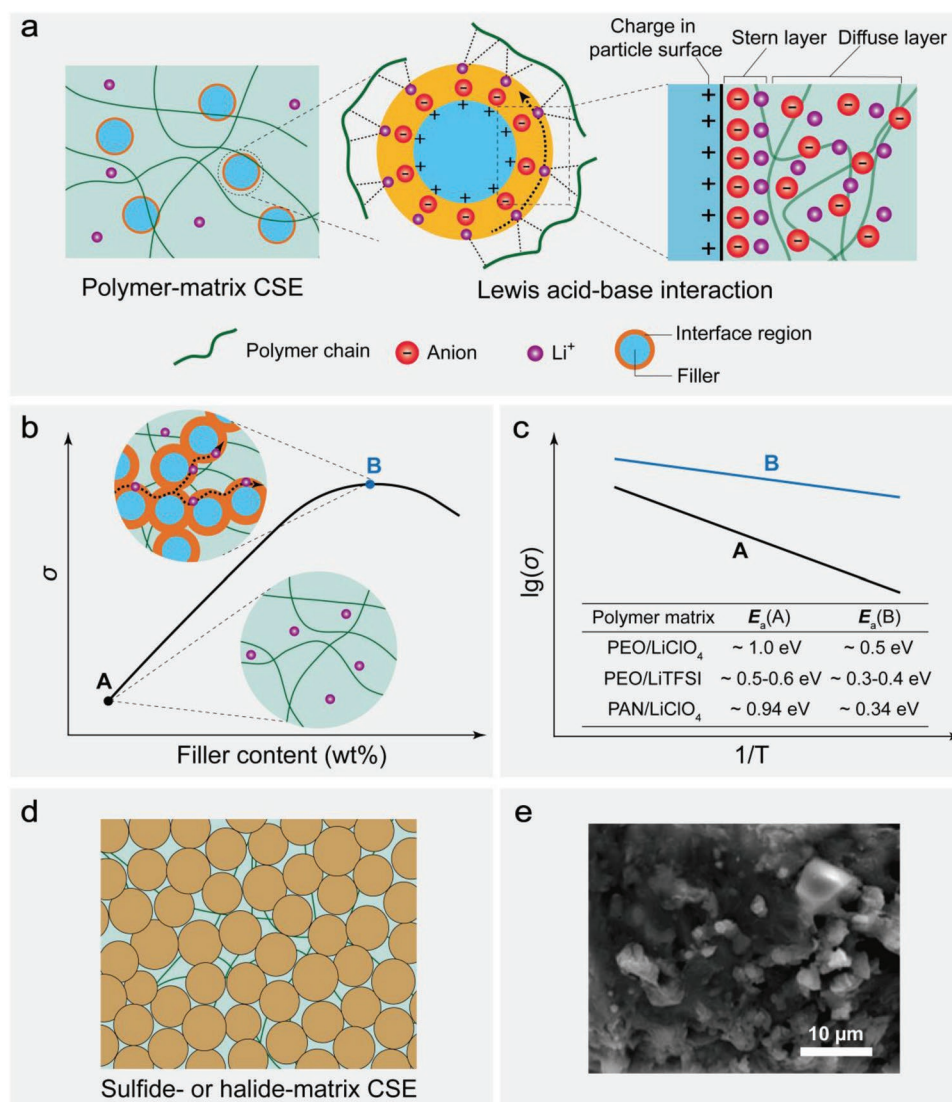


Figure 3. a) Schematic illustration of the polymer-matrix CSE. Lewis acid–base interactions between fillers and the polymer-Li salt system are shown in the two enlarged images at the right panel. Schematic dependency of ionic conductivity σ on b) the filler content and c) temperature. At Point A, there is no filler. At Point B, continuous percolation pathways along the interface layer of the fillers are formed. The activation energy E_a of the polymer-matrix CSEs is reduced by adding inorganic fillers due to the percolative pathways along the interface layer for ion transport. Typical E_a values of PEO/ LiClO_4 -matrix CSEs with oxide nanoparticles,^[70,71] PAN/ LiClO_4 -matrix CSE with oxide nanowires,^[75] and PEO/LiTFSI-matrix CSEs with oxide nanowires,^[76] are presented. PAN and LiTFSI represent polyacrylonitrile and bis(trifluoromethane)sulfonamide lithium, respectively. d) Schematic illustration of the sulfide- or halide-matrix CSEs. e) An SEM image of a $\text{Li}_6\text{PS}_5\text{Cl}$ -matrix CSE with poly(vinylidene fluoride-co-trifluoroethylene) as the binder. e) Reproduced with permission.^[82] Copyright 2022, Wiley-VCH.

Compared with nanoparticles, the fillers with high aspect ratios, such as nanofibers, nanowires, and nanosheets, would more easily generate a percolation network of interfacial layers.^[75,76] For example, compared with the granular LLTO filler, the fibrous LLTO fillers in the LLTO-polyacrylonitrile CSEs would generate continuous percolation interfaces that are more favorable for Li^+ transport due to the longer, more ordered ion-transport pathways and the absence of aggregation.^[75] When 2D vermiculite sheets with negatively charged surfaces were added to the PEO matrix, interfaces that effectively promote Li^+ transport can be formed, and the ionic conductivity was increased by about two orders of magnitude.^[77]

Regulating interactions between the fillers and the polymer matrix can improve the interfacial ion transport of the CSEs. Introducing favorable bonding between fillers and polymer matrix is an effective way.^[78,79] For example, a chemical bonding interaction between $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ and polyethylene glycol (PEG) was realized by in situ coupling reaction with a commercialized silane coupling agent, (3-chloropropyl)trimethoxysilane, which improved the interfacial compatibility between $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ and PEG. The PEG-matrix CSE film with $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ demonstrated an ionic conductivity of $9.83 \times 10^{-4} \text{ S cm}^{-1}$ at RT and a high Li^+ transference numbers of 0.68.^[78] Additionally, the defects or functional groups on the inorganic fillers can be utilized to

regulate the interactions too. For example, the flyash particles with abundant oxygen vacancies on the surface can efficiently immobilize (trifluoromethanesulfonyl) imide anions (TFSI[−]) and mitigate their irreversible oxidative decomposition due to the strong binding interaction between TFSI[−] anions and flyash particles. As a result, the introduction of flyash into the PEO-matrix CSEs improved both the Li⁺ transference number and electrochemical stability window (ESW).^[80] MXene-based mesoporous silica (MXene-mSiO₂) nanosheets provided a large quantity of functional groups (including −F from MXene and −OH from silica), which assisted with the generation of not only numerous Lewis acid–base interactions between MXene-mSiO₂ and Li salt, but also hydrogen bonds with the polymer matrix. Consequently, fast Li⁺ transport was enabled at the mesoporous nanosheets/polymer interface layers, leading to a high ionic conductivity of 4.6×10^{-4} S cm^{−1}, and the mechanical strength of the poly(propylene oxide) elastomer-matrix CSE was enhanced as well.^[81]

In sulfide- and halide-matrix CSEs, the polymer only acts as the binder to enable a thin, mechanically stable CSE membrane, and inorganic SE particles as the dominant component are closely packed together to form a 3D continuous network (Figure 3d,e).^[82] Although the ion transport happens primarily through the 3D continuous SE particle network, instead of the interface, regulating the interaction between the inorganic SE particles and the polymer is still necessary to balance the mechanical and electrochemical properties of the CSEs. The contact between particles determines the ionic conductivity of the CSEs.^[83,84] The intimate contact without the interface barriers arising from polymer binder layers or void gaps is critical. In comparison with sulfides and halides, oxides do not seem the ideal choice for the inorganic SE component in the kind of CSE membranes discussed above, because the oxide electrolyte particles are nondeformable and cannot easily form the intimate contact needed for efficient ion transport. As a result, more binder is usually needed, which significantly reduces the ionic conductivity of the resulting CSE membrane.

3. Interfaces within Composite Electrodes

The composite electrodes of SSBs typically consist of a reasonably high mass loading of active material particles (AMPs) that host Li-ions, a small number of electronic conductive agents, and a certain quantity of SEs (Figure 4a,b). The coexistence of SEs and AMPs in the composite electrodes produces a tremendous number of solid–solid AMP/SE interfaces, which might generate large charge transfer impedance due to the poor physical contact, (electro)chemical reactions, and the electrochemical–mechanical failure caused by the volume change of the AMPs.^[12–19,32,85–89] To address these issues, the interfaces in the composite electrodes need to be controlled rationally.

3.1. Physical Contact at Interfaces

The physical contact between AMPs and SEs in the composite electrodes is critical to the cycling stability and rate performance of SSBs. Generally, the SE particles are stochastically

distributed inside the composite electrodes to form a percolative Li-ion transport network around the AMPs. The presence of voids caused by the loose particle packing hinders the effective diffusion of ions between AMPs and SEs, leading to large interfacial resistance that compromises the electrochemical performance of SSBs (Figure 4a).^[90] More importantly, during the charge/discharge cycling, the significant volume change arising either from extraction/insertion of ions from/into intercalation-type AMPs or from phase transformation of conversion-type AMPs can cause large local strains. With most of its volume occupied by the nondeformable AMPs, the composite electrode is also rigid. Consequently, the strain at the interfaces between AMPs and SEs could easily lead to mechanical failures such as crack formation and delamination (Figure 4a). The associated increase of interfacial impedance would lower the electrochemical performance and even lead to battery failure.^[90–94] Furthermore, the sustained occurrence of ion-transport kinetic nonequilibrium at the solid–solid interfaces between AMPs and SEs could also result in the stress heterogeneity and the generation of microcracks within polycrystalline AMPs, leading to the deterioration of interfacial contact and the performance decay of SSBs.^[95] To realize and maintain the intimate contact between AMPs and SEs, the latter needs to possess the capability of assisting with the densification of the composite electrodes without high-temperature processing and accommodating the large volume change from the AMPs during cycling.

The composite electrodes of SSBs can be fabricated using either slurry mixing, tape casting, and pressing, or mixing and dry-coating/pressing. The flexibility and softness of polymer-matrix SEs allow for a relatively facile preparation of the composite electrodes.^[12] As a result, such materials not only could ensure low interfacial resistance between the AMPs and the SEs, but also are compatible with large-scale manufacturing processes. However, the elasticity of the polymer-matrix SEs is usually poor. Therefore, they undergo plastic deformation when the volume change of the AMPs occurs during cycling or when the electrode is subjected to an impact, resulting in the separation of the electrolyte from the AMPs and the resistance increase (Figure 4b).^[92,96,97] Developing a polymer-matrix SE with robust mechanical properties to accommodate the volume change is a promising way for maintaining good interface contact. A poly(propylene oxide) elastomer with a high resilience of 0.32 MJ cm^{−3} and an ionic conductivity of 0.25 mS cm^{−1} at RT was synthesized and used in a LiFePO₄ composite cathode. The elastomer with high elasticity worked as a robust binder and tolerated the repeated volume change of the LiFePO₄ particles; therefore, the integrity of the LiFePO₄/elastomer interface was elastically maintained during cycling, which in turn allowed for the stable operation of the SSB for 300 cycles with a capacity retention of 95% at 0.2 C.^[92] Therefore, developing polymer-matrix SEs that can serve as both the binder and the ionically conductive agent is a promising strategy. If the polymer is electronically conductive in the meantime, further performance improvement may be expected. For example, an electronically and ionically dual conductive polymer with a high elastic modulus of 6.8 GPa was coated on the AMPs, which assisted with the maintenance of the structural integrity of the cathode during cycling via buffering the internal stress. Accordingly, the SSBs with a high loading and high mass ratio of active

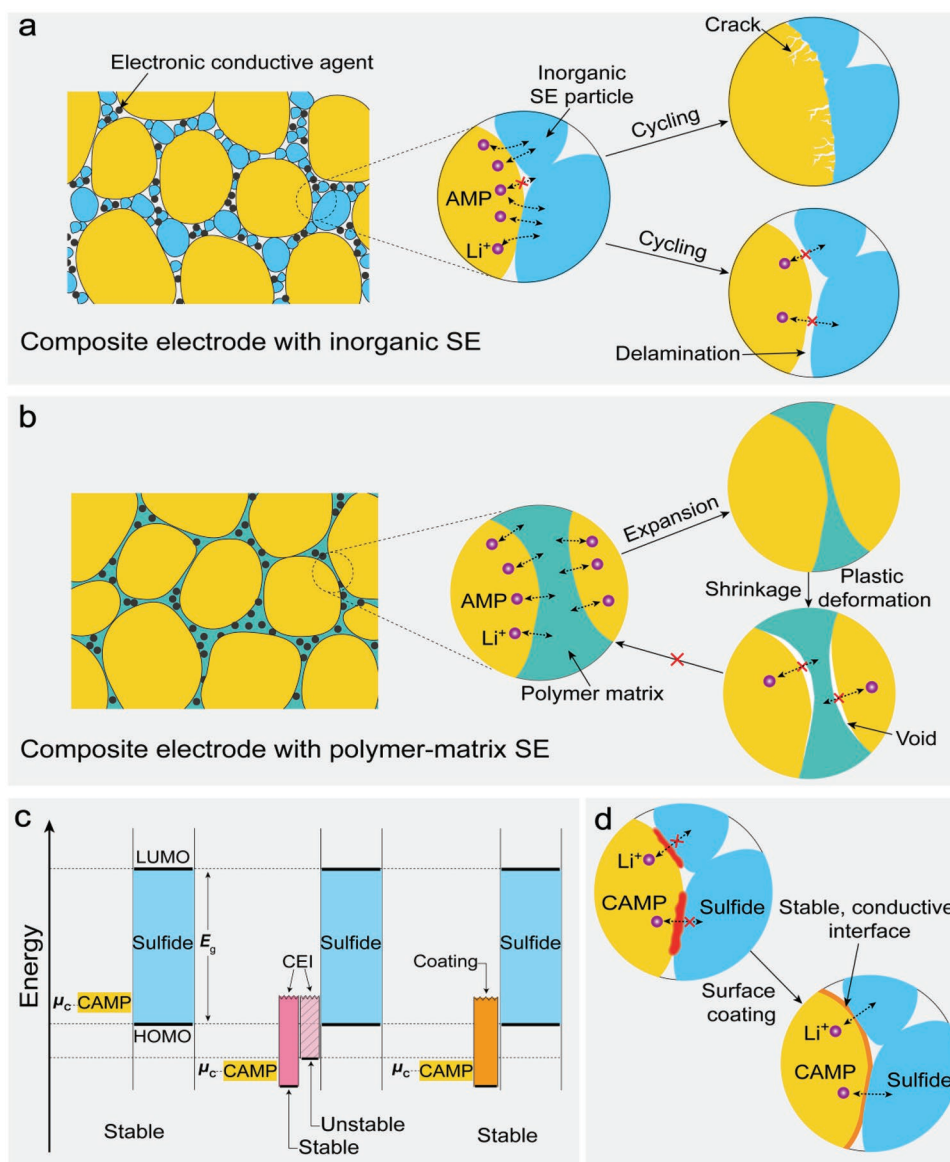


Figure 4. Microstructure of the composite electrodes with a) inorganic SE and b) polymer-matrix SE. In (a), voids appear between the inorganic SE particles and the AMPs. Cracking or delamination may occur due to the volume change of AMPs during cycling. In (b), intimate contact is realized between the polymer-matrix SE and the AMPs. Voids may appear after cycling because of the plastic deformation of the polymer-matrix SE. c) Schematic energy diagram of the CAMP and sulfide SE. When the μ_c of the CAMP is higher than the HOMO of the sulfide SE, the sulfide SE is not oxidized and the CAMP/sulfide system is thermodynamically stable (left). If the μ_c of the CAMP is lower than the HOMO of the sulfide SE, a cathode electrolyte interphase (CEI) with a HOMO lower than the μ_c is required to keep the CAMP/sulfide system stable (middle). If the HOMO of the CEI is higher than the μ_c , continuous interfacial reactions between the CAMP and the sulfide SE would occur and the CAMP/sulfide system is unstable (middle). When the μ_c of the CAMP is higher than the HOMO of the coating material, the CAMP/sulfide system is also stable (right). d) Schematic illustration of a stable, conductive interface established by using a surface coating on CAMPS.

materials showed a high Coulombic efficiency and excellent cycling stability.^[96]

When deformable inorganic SEs such as sulfides and halides are used in the composite cathodes, good physical contact between solid particles can be achieved via densification under pressure, and a suitable processing temperature may further improve the interfacial contact. It has been demonstrated that cold pressing at RT can create an intimate contact between AMPs and sulfide/halide SEs due to the plastic deformation of the SE particles.^[39,98] Table 1 shows some examples of the SSB

cells based on sulfide and halide SEs operating under different pressure.^[40,82,99–112] The cold pressing also avoids the potential atomic interdiffusion or side reactions at high temperatures. Nevertheless, if these undesired phenomena can be avoided, elevated temperatures can further promote the densification of composite cathodes.^[102] At such temperatures, the deformability of the sulfide and halide SEs would increase, and thus favor the formation of an intimate solid–solid contact. As shown in Table 1, e.g., the SSB cell with a $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622)– $(\text{Li}_2\text{S})_8(\text{P}_2\text{S}_5)_2(\text{Ni}_3\text{S}_2)_1$ composite cathode prepared by

Table 1. Examples of cycling performance of the SSB cells based on sulfide and halide SEs under different pressure.

SE	Anode cathode	Pressure to prepare cathode [MPa]	Testing conditions (T/P [MPa]) ^{a)}	Rate/current density	Life (cycle)	Capacity retention	Ref.
Li ₆ PS ₅ Cl	Li–In S	150	RT/150	0.1 C/0.176 mA cm ^{−2}	50	75%	[99]
Li ₆ PS ₅ Cl	Li–In LiNi _{1/3} Mn _{1/3} Co _{1/3} O ₂	100	RT/50	1/3 C/≈53 mA g ^{−1}	850	91.5%	[100]
Li ₆ PS ₅ Cl	Li LiNi _{0.80} Co _{0.15} Al _{0.05} O ₂	370	RT/5	0.1 C/–	100	80.9%	[101]
Li ₆ PS ₅ Cl	Li–In LiNi _{0.80} Co _{0.15} Al _{0.05} O ₂	50	RT/25	0.1 C/–	40	Low	[40]
		370				High	
(Li ₂ S) ₈ (P ₂ S ₅) ₂ (Ni ₃ S ₂) ₁	In LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂	500 (CP ^{b)})	–	0.1 C/18 mA g ^{−1}	50	61.5%	[102]
		500 (MP ^{c)})				91.7%	
Li ₆ PS ₅ Cl	Li–In LiCoO ₂	340	25 °C/–	0.1 C/16 mA g ^{−1}	100	93%	[103]
Li ₆ PS ₅ Cl	Li–In LiNi _{0.5} Co _{0.3} Mn _{0.2} O ₂	150	25 °C/100	0.1 C/15.5 mA g ^{−1}	200	80%	[104]
Li ₆ PS ₅ Cl	Li–In LiNi _{0.8} Co _{0.1} Mn _{0.1} O ₂	300	RT/50	0.1 C/0.126 mA cm ^{−2}	600	90%	[105]
Li _{5.5} PS _{4.5} Cl _{1.5}	Li–In LiNi _{0.7} Co _{0.1} Mn _{0.2} O ₂	380	60 °C/–	5 C/900 mA g ^{−1}	300	88%	[106]
Li ₁₀ GeP ₂ S ₁₂	In LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂	380	25 °C/–	0.5 C/80 mA g ^{−1}	500	72.7%	[107]
Li ₁₀ SnP ₂ S ₁₂	Li–In LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂	380	25 °C/100	0.1 C/–	100	87.6%	[108]
Li ₁₀ SnP ₂ S ₁₂	Li ₄ Ti ₅ O ₁₂ LiNi _{0.8} Co _{0.1} Mn _{0.1} O ₂	510	30 °C/25	0.3 C/54 mA g ^{−1}	100	79.7%	[109]
Li ₆ PS ₅ Cl ^{d)}	Li–In LiNi _{0.8} Co _{0.1} Mn _{0.1} O ₂	100	RT/100	1.61 C/1.0 mA cm ^{−2}	20 000	71%	[82]
Li ₃ InCl ₆	Li–In LiNi _{0.83} Mn _{0.06} Co _{0.11} O ₂	500	80 °C/2	–/1 mA cm ^{−2}	50	94%	[110]
Li ₃ InCl ₆	Li–In LiCoO ₂	360	–/70–80	0.2 C/0.16–0.19 mA cm ^{−2}	70	88%	[111]
Li ₃ YCl ₆	Li–In LiNi _{0.88} Co _{0.11} Al _{0.01} O ₂	370	30 °C/70	0.5 C/90 mA g ^{−1}	200	96.8%	[112]

^{a)}T and P represent temperature and pressure, respectively; ^{b)}CP represents cold pressing; ^{c)}MP means that the SE layer and the composite electrode are additionally pressed at 85 °C under 500 MPa after the CP process; ^{d)}The SE layer is a CSE made of Li₆PS₅Cl and poly(vinylidene fluoride-co-trifluoroethylene).

cold pressing and additional mild-temperature (85 °C) pressing showed an initial capacity of 103 mA h g^{−1}, larger than that of the SSB prepared by cold pressing (81 mA h g^{−1}); besides, a capacity retention of 91.7% was achieved after 50 cycles.^[102] In addition, the physical contact between silicon (Si)-based AMPs and sulfide SEs can be further improved by coating a conformal Li₆PS₅Cl electrolyte layer on the AMPs. Such a Li-ion conductive coating provided an intimate interface with the surrounding SE for the AMPs, effectively reduced the amount of SE in the composite anode, and enabled the increase of the energy density.^[113]

3.2. Chemical and Electrochemical Stability at Interfaces

In addition to the intimate physical contact and large contact area, it is also necessary to ensure the (electro)chemical stability between the cathode active material particles (CAMPs) and the SE; otherwise, undesired side reactions or elemental interdiffusion may occur in the composite cathodes and deteriorate the cell performance. For example, the oxidation potential of LiCoO₂ is higher than the anodic limit of PEO-matrix SEs. As a result, they would react with each other in the composite cathode; the associated decomposition of PEO-matrix SEs and self-oxygen-release of LiCoO₂ lead to rapid degradation of the interface.^[114] Another example is the composite cathode formed by LiCoO₂ and sulfide SEs. The mutual diffusions of Co, P, and S may occur at the LiCoO₂/Li₂S–P₂S₅ interface after initial charging, resulting in an undesired interfacial layer.^[115] In comparison, the halide SEs exhibit electrochemical stability against

oxide cathode materials, and thus do not require any extra surface coating on the CAMPs.^[19,38,39,116] Regardless, such SEs are rare. In most cases, the interfaces still need optimization. To this end, extensive attention has been paid on the modification of interfacial chemistry in the composite cathodes by using surface coatings or highly stable catholytes.

To avoid adverse interfacial reactions between the CAMPs and sulfide SEs, the surface of CAMPs is often coated with a very thin layer of functional oxides. When the chemical potential μ_c of the CAMPs is higher than the highest occupied molecular orbital (HOMO) of the coating material, the CAMP/sulfide system is thermodynamically stable (Figure 4c); therefore, the chemical reaction between CAMPs and sulfide SEs is prevented and a stable, conductive interface can be established (Figure 4d).^[88,117,118] Ideally, the surface coating on the CAMPs should have the following features: 1) good uniformity for completely isolating the CAMPs from SEs; 2) very small thickness for minimizing charge transfer resistance; 3) high ionic conductivity; 4) excellent tolerance to the volume change of CAMPs during discharge/charge; 5) good chemical and electrochemical stability to avoid harmful side reactions; 6) simple and scalable synthetic processes for mass production.^[119–123] As shown in Table 2, various oxide coatings such as Li₂WO₄,^[103] LiNbO₃,^[30,82,128] and LLTO,^[104,125] have been used. For example, a LiNbO₃ coating layer with about 5 nm in thickness can effectively improve the interfacial stability between the LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM811) cathode and Li₁₀GeP₂S₁₂ sulfide. Consequently, a solid-state Li–In||Li₁₀GeP₂S₁₂||LiNbO₃-coated NCM811 cell displayed a high discharge capacity of 203 mA h g^{−1} at 0.1 C and good rate performance (136.8 mA h g^{−1} at 5 C) at

Table 2. Some coatings on the CAMPs used in the SSB cells based on sulfide SEs.

Cathode	Coating	SE in composite cathode	Anode	Testing T [°C]	Voltage [V]	Rate/current density	Life (cycle)	Capacity retention	Ref.
LiCoO ₂	LiNbO ₃	Li _{9.54} Si _{1.74} P _{1.44} S _{11.7} Cl _{0.3}	Li ₄ Ti ₅ O ₁₂	100	1.0–2.6	18 C/12 mA cm ⁻²	500	75%	[30]
LiCoO ₂	Li ₂ WO ₄	Li ₆ PS ₅ Cl	Li–In	25	2.2–3.6	0.1 C/16 mA g ⁻¹	100	93%	[103]
LiCoO ₂	Li ₂ CoTi ₃ O ₈	Li ₁₀ GeP ₂ S ₁₂	Li–In	25	1.5–3.7	0.1 C/15 mA g ⁻¹	200	82.9%	[124]
LiNi _{1/3} Mn _{1/3} Co _{1/3} O ₂	Li _{0.35} La _{0.5} Sr _{0.05} TiO ₃	Li ₆ PS ₅ Cl	Li–In	RT	2.5–4.0	1/3 C/53 mA g ⁻¹	850	91.5%	[100]
LiNi _{0.5} Co _{0.3} Mn _{0.2} O ₂	Li _{0.5} La _{0.5} TiO ₃	Li ₆ PS ₅ Cl	Li–In	25	2.3–3.8	0.1 C/15.5 mA g ⁻¹	200	80%	[104]
LiNi _{0.7} Co _{0.1} Mn _{0.2} O ₂	LiNbO ₃	Li _{5.5} PS _{4.5} Cl _{1.5}	Li–In	60	2.4–3.7	5 C/900 mA g ⁻¹	300	88%	[106]
S ^a -NCM622	Li _{0.35} La _{0.5} TiO ₃	Li ₆ PS ₅ Cl	Li–In	RT	2.2–3.7	0.1 C/20 mA g ⁻¹	100	84.5%	[125]
S-NCM622	Li _{1.4} Al _{0.4} Ti _{1.6} (PO ₄) ₃ /cyclized PAN	Li ₁₀ GeP ₂ S ₁₂	In	25	1.9–3.8	0.5 C/80 mA g ⁻¹	500	72.7%	[107]
S-NCM622	Li _{1.4} Al _{0.4} Ti _{1.6} (PO ₄) ₃	Li ₁₀ SnP ₂ S ₁₂	Li–In	25	2.2–3.7	0.1 C/–	100	87.6%	[108]
NCM622	Li ₂ CO ₃ /LiNbO ₃	β-Li ₃ PS ₄	Li ₄ Ti ₅ O ₁₂	25	1.35–2.85	0.1 C/18 mA g ⁻¹	100	91%	[126]
Ti-doped S-NCM622	TiNb ₂ O ₇	Li ₁₀ GeP ₂ S ₁₂	Li–In	RT	2.72–4.4 ^b	0.1 C/18 mA g ⁻¹	140	92.2%	[127]
LiNi _{0.80} Co _{0.15} Al _{0.05} O ₂	LiNbO ₃	Li ₆ PS ₅ Cl	Li	RT	2.5–4.3	0.1 C/–	100	80.9%	[101]
NCM811	LiNbO ₃	Li ₆ PS ₅ Cl	Li–In	RT	2.2–3.6	1.61 C/1 mA cm ⁻²	20 000	71%	[82]
NCM811	LiNbO ₃	Li ₁₀ GeP ₂ S ₁₂	Li–In	35	2.1–3.78	0.5 C/100 mA g ⁻¹	50	77.9%	[128]
S-NCM811	LiNbO ₃	Li ₁₀ SnP ₂ S ₁₂	Li ₄ Ti ₅ O ₁₂	30	2.85–4.35 ^b	0.3 C/54 mA g ⁻¹	400	67.3%	[129]
S-NCM811	LiNbO ₃	Li ₁₀ SnP ₂ S ₁₂	Li ₄ Ti ₅ O ₁₂	30	2.85–4.35 ^b	0.3 C/54 mA g ⁻¹	100	79.7%	[109]
NCM811	LiOH	Li ₆ PS ₅ Cl	Li–In	RT	2.5–4.2	0.1 C/0.126 mA cm ⁻²	600	90%	[105]
NCM811@LiCoO ₂	LiNbO ₃	Li ₁₀ GeP ₂ S ₁₂	Li–In	35	2.1–3.78	0.3 C/0.612 mA cm ⁻²	585	80%	[130]
LiNi _{0.82} Co _{0.12} Mn _{0.06} O ₂	LiTaO ₃	Li ₆ PS ₅ Cl	Li–In	30	1.9–3.9	0.2 C/34 mA g ⁻¹	30	83%	[131]

^a)S represents single crystal; ^b)Voltage versus Li/Li⁺.

60 °C.^[128] A solid-state cell with a composite cathode made of LiNbO₃-coated NCM811 and the Li₆PS₅Cl electrolyte demonstrated an ultralong cycle life of 20 000 cycles with a current density of 1.0 mA cm⁻² at RT.^[82] A crystalline LLTO coating layer with a thickness of ≈6 nm and a high Li-ion conductivity of 0.3 mS cm⁻¹ was also adopted to improve the structural stability of the single-crystalline NCM622 particles and to solve the interfacial problems (such as the interfacial reaction and the space charge layer) between NCM622 and the Li₆PS₅Cl electrolyte.^[125]

When polymer SEs are used as the catholytes in the composite cathodes, it is necessary to improve the ESW of the polymer SEs and maintain the interfacial stability between the CAMPs and the catholytes.^[132] The introduction of LLZO particles into PEO would result in an improved oxidation potential up to ≈5 V. When this solid catholyte was used in an NCM622 composite cathode, the SSB was successfully cycled at 70 °C with a high cut-off potential of 4.2 V (vs Li/Li⁺) and delivered an average discharge capacity of 160 mA h g⁻¹ at C/20.^[133] Replacing the PEO-matrix catholyte with a poly(propylene carbonate)-matrix one can also result in a remarkably improved cycling performance for an SSB with an NCM622 cathode, because the catholyte oxidation was suppressed and the cathode resistance was reduced due to the excellent high-voltage stability of poly(propylene carbonate).^[134] Besides, it should be noted that the electronically conductive carbon in the composite cathode might accelerate the decomposition of PEO-matrix SEs at high voltages.^[135–137]

4. Interfaces between Electrodes and Solid Electrolyte Separators

On a par with the internal interfaces within the SEs and composite electrodes, the interfaces between electrodes and SE separators, namely, the cathode/SE separator or anode/SE separator interfaces (Figure 5a), also play vital roles in Li-ion transport and the cycling performance of SSBs. In this section, we will discuss the issues of these interfaces, mainly concerning physical contact, (electro)chemical stability, and Li dendrite growth, and some strategies.

4.1. Interface between Cathodes and Solid Electrolyte Separators

In the assembly of SSBs, the fabrication processes such as calendering or pressing are often applied to improve the contact between the cathodes and the SE separators. There are several challenges for these interfaces, such as poor interfacial contact and undesired side reactions. These interfacial issues may lead to large resistance and sluggish ion migration (Figure 5a).^[138,139]

Similar to the composite electrodes, polymer-matrix electrolytes with good elasticity are more beneficial for maintaining intimate solid–solid contact, which is crucial for achieving low interfacial resistance and satisfactory cycling stability.^[197,140] Various strategies have been proposed to improve the interfaces between the cathodes and the polymer-matrix SEs. Among the reported approaches, in situ polymerization is an effective one.

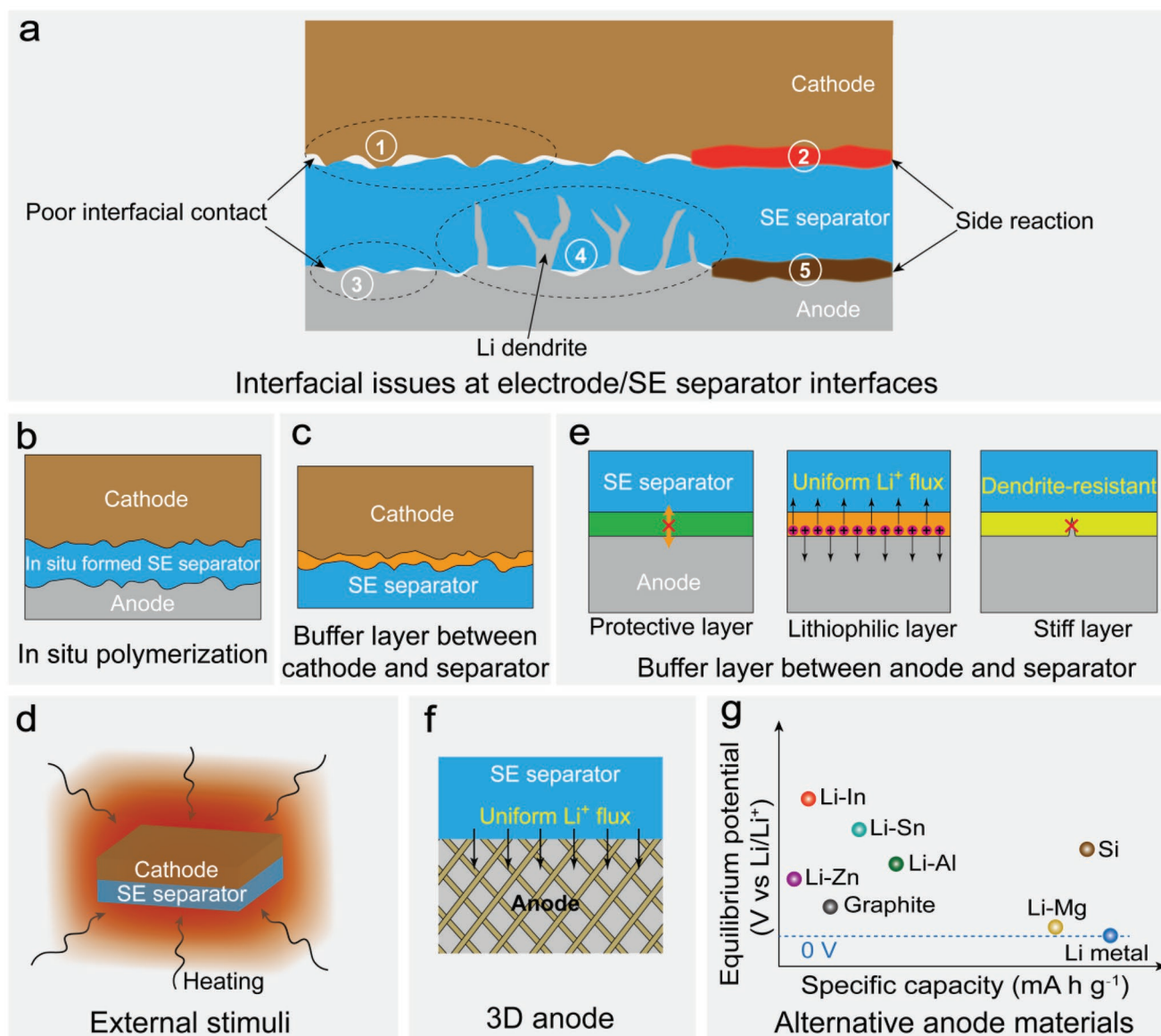


Figure 5. a) Interfacial issues (marked with the numbers of ①–⑤) at the electrode/SE separator interfaces and b–g) corresponding solutions. b) In situ polymerization of SEs (solution for ①, ③, and ④). c) Building a buffer layer between the cathode and the SE separator (solution for ① and ②). d) External stimuli used to bond the cathode and the SE separator (solution for ①). e) Building a buffer layer between the anode and the SE separator. The buffer layer can be a protective layer to prevent side reactions (solution for ⑤), a lithiophilic layer to improve the wettability of Li metal with the SE separator (solution for ③ and ④), or a stiff layer to suppress Li dendrite growth (solution for ④). f) 3D structural design of anodes (solution for ③ and ④). g) Alternative anode materials with different equilibrium potentials (solution for ④ and ⑤).^[8–19,218–223]

The polymer-matrix SEs synthesized by in situ polymerization can wet the electrode surface and promote interfacial Li-ion transport (Figure 5b).^[140,141] For example, a polymer SE was synthesized by ring-opening polymerization of 1,3-dioxolane (DOL) inside an electrochemical cell, which was able to wet both the cathode and anode and reduce the interfacial resistance. The structure of the poly-DOL SE was unchanged during cycling, resulting in high Coulombic efficiencies (close to 100%) and excellent reversibility for the $\text{Li|poly-DOL SE|LiFePO}_4$ SSB cells even after 700 cycles.^[140] Additionally, the in situ polymerization of poly(ethylene glycol) methyl ether acrylate (PEGMEA) between a self-supported 3D porous $\text{Li}_6\text{PS}_5\text{Cl}$ framework and a cathode may realize effective interfaces. The soft, in situ

polymerized PEGMEA could eliminate the voids both within the cathode and at the cathode/ $\text{Li}_6\text{PS}_5\text{Cl}$ separator interface, while the $\text{Li}_6\text{PS}_5\text{Cl}$ framework provided fast ion-transport paths. Therefore, efficient ion transport was ensured and the resistance of the SSB cell was stable during cycling. An NCM811-based SSB cell with the PEGMEA- $\text{Li}_6\text{PS}_5\text{Cl}$ composite electrolyte exhibited a high initial discharge of 193 mA h g^{-1} and a high Coulombic efficiency of more than 99% during subsequent cycles at RT.^[141]

Introducing a mechanically deformable and (electro)chemically stable material as the buffer layer between the cathode and the SE separator is also an attractive strategy (Figure 5c), as it not only ameliorates poor physical contact at the interface, but

also avoids the direct exposure of chemically/electrochemically unstable SEs to CAMPs.^[142] The buffer layers should exhibit good wettability with both cathodes and SE separators, strong adhesiveness (to ensure an intimate contact between cathodes and SE separators during cycling), high ionic conductivity, and chemical/electrochemical stability toward both cathodes and SE separators.^[143–146] A high-voltage stable solid-state buffer layer based on polyfluoroalkyl acrylate was constructed via in situ solvent-free bulk electro-polymerization between an NCM811-based cathode and a PEO-matrix SE.^[144] The buffer layer widened the ESW to 5.1 V (vs Li/Li⁺), which was much broader than that of the PEO matrix, and achieved a high ionic conductivity of 1.02×10^{-4} S cm⁻¹ at RT. This buffer layer has advantages of excellent wettability and strong capability of protecting PEO, rendering intimate physical contact to the cathode/SE separator interface and substantially inhibiting the space charge effect. This in situ electrochemical process is a promising interface engineering method for the large-scale production of SSBs.^[144] Besides in situ buffer layers, ex situ buffer layers have also been used.^[145,146] A poly(*N*-methyl-malonic amide) (PMA)-based buffer layer was inserted between a PEO-matrix SE separator and a LiCoO₂-based cathode to protect the PEO from a high-voltage oxidation. In this way, a stable polymer/oxide interface was achieved, and the Li⁺ transport at the PEO/PMA interface was not impeded.^[145]

Pure oxide ceramic electrolytes were also studied as separators.^[15,16,18] To improve the interface between the brittle ceramic separators and the composite cathodes, buffer layers are required as well (Figure 5c).^[143,147] For example, a curable polymer-matrix glue electrolyte was applied between a sulfurized polyacrylonitrile-matrix composite cathode and an LLZO-based ceramic separator, aiming to address the poor solid–solid contact and thus decrease the high interfacial resistance. With the glue modification, a solid-state Li-S cell showed a capacity retention of 95.2% after 50 cycles at RT and no obvious failure was observed at the cathode/separator interface.^[143] Also, external stimuli such as electrical thermal annealing and microwave heating are feasible to tightly bond the cathodes and oxide SE separators (Figure 5d).^[148,149] An intimate interfacial contact was formed between a V₂O₅ cathode and an LLZO-based ceramic separator by electrical thermal annealing, leading to a significant decrease of the interfacial impedance.^[148] Microwave soldering with a feature of selective heating was able to rapidly melt and re-solidify the surface of V₂O₅ cathode particles within a few seconds, resulting in a strong bonding between the V₂O₅ cathode and the LLZO ceramic separator along with a 28-fold decrease in the interfacial resistance.^[149] In addition, 3D structural designs of the cathodes and SEs may be implemented to increase the contact area between the cathodes and the oxide ceramic separators.^[150–152]

Although several methods have been used to improve the interface between the brittle oxide ceramic separators and the composite cathodes, it is difficult for the ceramic electrolyte separators to buffer the volume change of the electrode materials during long-term cycling due to their rigidity. Consequently, mechanical failures such as crack formation and delamination at the cathode/electrolyte interfaces are inevitable. Furthermore, it is hard to fabricate bulk SSBs based on oxide ceramic separators because of the high processing temperature and the

intrinsic brittleness of oxides. Instead, such separators might be applicable for micro-batteries fabricated by thin-film deposition. Alternatively, they might also be suitable for the multilayered ceramic batteries fabricated by low-temperature co-firing processing that barely induces atomic interdiffusion or reactions between CAMPs and oxide electrolytes, but the manufacture of such batteries is challenging.^[12,15,153]

4.2. Interfaces between Anodes and Solid Electrolyte Separators

For the interfaces between the SE separators and the anodes, the major issues are the interfacial reactions and Li dendrite growth (Figure 5a). These issues are especially severe when Li metal is directly used as the anode. Essentially, the Li stripping/plating behaviors in the SSBs with a Li metal anode are the processes involving the Li-ion transport and Li deposition at the Li anode/SE separator interfaces. Deeply understanding the stability, kinetics, and failure mechanism of the interfaces is of vital significance toward the practical applications of the Li-metal anode.

4.2.1. Chemical/Electrochemical Stability

The chemical/electrochemical stability of the Li/SE interfaces is a challenge for high-energy-density SSBs with a Li-metal anode. The interfaces between Li metal and the SE separators can be divided into three categories based on the physical and (electro) chemical characteristics: thermodynamically stable interface, mixed electronically and ionically conducting interphase (MCI), and ionically conductive but electronically insulating interphase similar to solid electrolyte interphase (SEI) in liquid electrolytes.^[8,154] Thermodynamically stable interface refers to the situation that SEs are chemically inert to Li metal without the formation of any new interphase. This type of interface is most desired, but is extremely rare. The other two are thermodynamically unstable interfaces accompanied by the formation of interphases. Since the MCI, i.e., the second type of interface, can transport both Li-ions and electrons, it would allow the anode to continuously lithiate and reduce the unreacted SEs. The growth of such an interphase would not stop until either the SE or the anode is completely consumed. Therefore, this type of interface behavior is completely unacceptable for SSB operation.^[118,155,156] In contrast, SEI, i.e., the third type of interface mentioned above, can transport ions but not electrons, thereby preventing the anode from further lithiating and reducing the SEs. If the SEI happens to exhibit a sufficiently high ionic conductivity, it can also enable decent cell performance.

The interfacial stability of various inorganic SEs against Li metal have been investigated.^[157,158] The materials that are thermodynamically stable with Li metal are mainly binary Li compounds, such as LiF, LiCl, LiBr, Li₂O, Li₂S, LiH, and Li₃N. However, the ionic conductivities of such materials are unacceptably low, so they may only serve as very thin protective layers for Li metal, instead of SEs. Beyond these compounds, most inorganic SEs would react with Li metal to form either MCI or SEI.^[32,88] This instability mainly arises from the high-valence cations (e.g., P⁵⁺, Ge⁴⁺, Ta⁵⁺, Ti⁴⁺, and In³⁺) that broadly

exist in LLZO, LLTO, NASICON-type electrolytes, sulfides, and halides.^[19,32,88,154,159–162] Although most of the mainstream inorganic SEs are unstable against Li, some of them would only undergo self-limited reaction upon contact with Li, and the resulting SEI would not considerably impede ion transport either. For example, although the direct contact between Al-doped LLZO and Li would induce a local structural phase transition of LLZO, the interphase is an electronically insulating but ionically conductive SEI with a thickness of less than 5 unit cells.^[58] Such a thin layer will not significantly hinder the overall ion transport, so an acceptably low interfacial resistance can be achieved. In addition to LLZO, certain P-containing sulfide SEs such as $\text{Li}_7\text{P}_3\text{S}_{11}$ and $\text{Li}_6\text{PS}_5\text{Cl}$ exhibit a benign interfacial behavior too, because none of their reduction products is highly electronically conductive.^[17] As for polymer SEs, many of them are not stable against Li either. When directly contacted with Li metal, PEO-matrix polymer electrolytes could be decomposed into C_2H_4 and H_2 during cycling and a Li_2O -containing interphase is formed, leading to the impedance increase and poor cycling performance of SSBs.^[163] Poly(vinylidene difluoride) (PVDF)-matrix polymer electrolytes could be dehydrofluorinated under the attack of the free radicals originating from the reaction between bound residual solvent and Li metal during cycling.^[164,165]

4.2.2. Li Dendrite Growth

The growth of Li dendrites in SEs is a long-standing serious issue; uncontrolled dendrite growth would extend into the electrolyte separator and eventually short-circuit the cells.^[166,167] The growth of Li dendrites is a result of electro-chemo-mechanical coupling, which makes it one of the most complex phenomena in SSBs.^[156] There are many factors causing the growth of Li dendrites. First of all, the dynamic difference between the Li stripping at the interface and the Li-atom diffusion from Li metal bulk would result in gradual formation of voids at the Li/SE interfaces, leading to a reduced interfacial contact area, severe polarization, and the increase of interfacial impedance.^[155,168–170] The reduction in the contact area brings about local current increase or the formation of hotspots of current density, which promotes the growth of dendrites and further accelerates interface failure.^[171] Additionally, an interfacial layer with atomic-level defects on Li metal is highly detrimental, as it will induce the emergence of interfacial pores and the contact loss during Li stripping, both of which facilitate the Li dendrite growth.^[172] Second, many SEs show poor wettability with Li metal. Consequently, it is difficult for these SEs to realize an intimate contact with Li metal, which increases the chance of Li dendrite growth.^[32,88,173] Third, the intrinsic properties of SEs may profoundly influence both the formation and the growth of Li dendrites. Although presently this issue has not been fully understood, the existing studies have definitively demonstrated that the electronic conductivity and elastic modulus are two critical factors. If the SE exhibits lower electronic conductivity and higher elastic modulus, the dendrites would be more difficult to form and grow.^[63,174–176] Last but not least, external experimental conditions such as pressure may also influence the Li dendrite growth. If the applied pressure is too high, the

SSBs can be shorted either immediately or after a relative short cycling time due to the creeping of ductile Li metal through the SEs.^[101]

Many strategies have been developed to provide solutions for the two issues of the Li/SE interfaces above. First of all, building a buffer layer is the most widely used strategy; the buffer layer may have multiple functions, such as avoiding the decomposition of SEs, improving the wettability of Li metal with SEs, and suppressing Li dendrite growth (Figure 5e). Various buffer layers are summarized in Table 3.^[164,177–202] A typical example for the application of this strategy is the buffer layer for sulfide SEs, which have high ionic conductivity but suffer from spontaneous reaction with Li.^[157,158] To realize satisfactory interface between sulfide SEs and Li, various buffer layers such as Ag,^[177] Li-argyrodites,^[178] Si_xSiS_y ,^[181] LiH_2PO_4 ,^[179] and plastic crystal electrolyte^[180] have been applied, as shown in Table 3. For instance, a LiH_2PO_4 protective layer was in situ formed on the surface of a Li metal anode to suppress the growth of the mixed ionic-electronic reactants into the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ sulfide electrolyte.^[179] The reaction between halide electrolytes and Li metal can also be prevented by using buffer layers. A thin $\text{Li}_6\text{PS}_5\text{Cl}$ argyrodite protection layer was used to avoid the direct physical contact between the Li anode and Li_3YCl_6 .^[183] Without the $\text{Li}_6\text{PS}_5\text{Cl}$ buffer layer, the interphase would form directly between Li and Li_3YCl_6 and grow continuously due to its mixed conducting nature. In comparison, the interphase between Li and $\text{Li}_6\text{PS}_5\text{Cl}$ is poorly electronically conductive, so its growth is self-limiting. In addition to sulfides and halides, the strategy of applying a buffer layer is also rather effective for oxide SEs, though the issues to be addressed are different. Oxides usually show poor wettability with Li metal, so it is necessary to make their surface lithiophilic; otherwise, a continuous face-to-face contact cannot be formed, and the poor contact, as mentioned above, will promote Li dendrites.^[204] Constructing a lithiophilic buffer layer on the oxide surface can effectively enhance the contact between Li metal and oxides. The buffer layer needs to possess the following characteristics: 1) good lithiophilicity; 2) (electro)chemical inertness with the SEs; 3) high Li-ion diffusivity and low thickness; 4) good stability during cycling.^[205,206] Various lithiophilic materials, including metals (e.g., Zn,^[185] Si ,^[207] Au,^[186] Al,^[187] Ge^[188,208]), oxides (e.g., Al_2O_3 ,^[189] ZnO ,^[18,190]), fluorides (e.g., SnF_2 ,^[191] LiF ,^[192]), nitrides (e.g., Li_3N ,^[193,209]), and polymers (e.g., PEO^[196]) have been applied, as shown in Table 3. For example, an ultrathin Al_2O_3 film (averagely 5–6 nm) was deposited on the surface of $\text{Li}_7\text{La}_{2.75}\text{Ca}_{0.25}\text{Zr}_{1.75}\text{Nb}_{0.25}\text{O}_{12}$ SE by atomic layer deposition. Such an ultrathin lithiated Al_2O_3 film ensured the proper wetting of metallic Li on the surface of the SE and allowed for effective Li^+ transport. As a result, the Li/SE interfacial impedance was remarkably decreased from 1710 to $1\ \Omega\ \text{cm}^2$.^[189] According to Monroe and Newman, the shear modulus of the buffer layer needs to be at least twice that of Li metal in order to suppress Li dendrite growth.^[210] Following this theory, a LiPON thin-film coating with a shear modulus of $\approx 31\ \text{GPa}$, which was nearly nine times that of Li metal ($\approx 3.4\ \text{GPa}$), was deposited on Li metal by magnetron sputtering, and no Li dendrite was observed in the SSB based on such a LiPON-protected Li anode during cycling.^[211]

Table 3. Some buffer layers used at the Li/SE interfaces.

SE	Buffer layer	Cycling conditions of Li Li cell (current density [mA cm^{-2}] & capacity [mA h cm^{-2}], T)	Duration [h]	Ref.
$\text{Li}_{10}\text{GeP}_2\text{S}_{12}$	Ag	0.1 & 1, 25 °C	1000	[177]
$\text{Li}_{10}\text{GeP}_2\text{S}_{12}$	$\text{Li}_{5.5}\text{PS}_{4.5}\text{Cl}_{1.5}$	0.25 & 0.25, RT	1800	[178]
$\text{Li}_{10}\text{GeP}_2\text{S}_{12}$	LiH_2PO_4	0.1 & 0.05, 25 °C	~ 950	[179]
$\text{Li}_{10}\text{GeP}_2\text{S}_{12}$	PCE ^{a)}	0.13 & 0.13, RT	250	[180]
Li_3PS_4	Li_xSiS_y	0.1 & 0.5, RT	2000	[181]
$\text{Li}_7\text{P}_3\text{S}_{11}$	LiF	0.5 & 0.1, 25	200	[182]
Li_3YCl_6	$\text{Li}_6\text{PS}_5\text{Cl}$	0.2 & 0.2, 25 °C	1000	[183]
$\text{Li}_3\text{YBr}_{5.7}\text{F}_{0.3}$	LiF/YF_x	0.1 & 0.1, –	1000	[184]
$\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$	Zn–Li–N–O ^{b)}	1 & 2, RT	5500	[185]
$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Ta}_{0.4}\text{O}_{12}$	Au	0.25 & 0.5, 50	150	[186]
LLZTO–LZO ^{c)}	Li–Al alloy	0.2 & –, 25	3000	[187]
$\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{0.5}\text{P}_3\text{O}_{12}$	Ge	0.3 & 0.3, RT	200	[188]
$\text{Li}_7\text{La}_{2.75}\text{Ca}_{0.25}\text{Zr}_{1.75}\text{Nb}_{0.25}\text{O}_{12}$	Al_2O_3	0.2 & 0.1, RT	90	[189]
$\text{Li}_{1.4}\text{Al}_{0.4}\text{Ti}_{1.6}(\text{PO}_4)_3$	ZnO	0.2 & 0.1, RT	1000	[190]
$\text{Li}_{6.75}\text{La}_3\text{Zr}_{1.75}\text{Ta}_{0.25}\text{O}_{12}$	SnF_2	0.05 & 0.025, 30 °C	500	[191]
$\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$	LiF	0.3 & 0.15, RT	600	[192]
$\text{Li}_7\text{La}_3\text{Zr}_5\text{O}_{12}$	Li_3N	0.3 & 0.15, RT	300	[193]
$\text{Li}_{5.9}\text{Al}_{0.2}\text{La}_3\text{Zr}_{1.75}\text{W}_{0.25}\text{O}_{12}$	graphite	0.3 & 0.3, RT	1000	[194]
$\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$	$\text{Li}_3\text{N/Cu}$	0.1 & 0.05, 25 °C	1000	[195]
$\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$	PEO/BN ^{d)}	0.3 & 0.15, 60 °C	500	[196]
PVDF/LiFSI ^{e)}	LiF	0.5 & 0.25, 25 °C	~200	[164]
PEO/LiTFSI– Al_2O_3 – $\text{Mg}(\text{ClO}_4)_2$	$\text{Li}_2\text{MgCl}_4/\text{LiF}$	1.2 & 0.1, 55 °C	100	[197]
PEO/LiTFSI–LLZTO ^{f)} – LiNO_3	$\text{Li}_3\text{N/LiF}$	0.2 & 0.2, 60 °C	800	[198]
PEO/LiTFSI– Li_2S	LiF	0.1 & 0.1, 50 °C	1800	[199]
PEO/LiTFSI–LPS ^{g)}	$\text{Li}_3\text{PS}_4/\text{Li}_2\text{S/LiF}$	0.2 & 0.4, 60 °C	3475	[200]
PVDF/LiTFSI–LLZTO ^{h)}	PHCE ⁱ⁾	1 & 0.5, RT	~430	[201]
PAN–PEO/LiTFSI	$\text{Li}_3\text{N/LiF}$	0.5 & 0.25, 60 °C	300	[202]

^{a)}PCE represents the plastic crystal electrolyte that is succinonitrile with 5% LiTFSI and 2% LiNO_3 ; ^{b)}Zn–Li–N–O represents a composite of Zn, ZnLi_x alloy, Li_3N , Li_2O , and other species; ^{c)}LLZTO–LZO represents $\text{La}_2\text{Zr}_2\text{O}_7$ densified $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$; ^{d)}BN represents boron nitride; ^{e)}LiFSI represents lithium bis(fluorosulfonyl)imide; ^{f)}LLZTO represents $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$; ^{g)}LPS represents the mixture of Li_2S , S, and P_2S_5 with a molar ratio of 1:2:1; ^{h)}LLZTO represents $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$; ⁱ⁾PHCE represents the ultraviolet-polymerized high concentrated electrolyte that is made of poly (ethylene glycol)methyl ether methacrylate, LiFSI, and bisphenol A ethoxylate dimethacrylate.

Besides the aforementioned artificial buffer layers, in situ formation of a stable buffer layer at the anode/SE separator interface is also an effective strategy to enhance the interfacial stability (Table 3). It has been found that the in situ-formed LiF/ Li_3N -enriched buffer layers can improve the Li-metal compatibilities of the polymer-, halide-, and sulfide-based SEs, because these interfacial layers improve the contact between the SEs and the Li metal anode, prevent continuous interfacial reactions, and homogenize the Li deposition.^[164,184,197–199,202,212,213] To in situ generate the LiF/ Li_3N -enriched buffer layers at Li/PEO-matrix polymer electrolytes, various additives such as LiNO_3 , Li_2S , and Li_2MgCl_4 have been introduced into the electrolytes to promote the decomposition of Li salts (Table 3).^[197–199] For example, a uniform buffer layer that was a Li^+ conductor and an electronic insulator can be in situ generated at the Li/PEO-electrolyte interface with the assistance of Li_2MgCl_4 , which contributed to the homogenization of the Li^+

flux during Li plating, the increase of the critical current density for Li dendrite growth, and the stable cycling of the SSBs based on LiFePO_4 and NCM811.^[197] The bound residual solvent in the PVDF-matrix SEs can also assist with the in situ formation of the buffer layer.^[164,165,214] The electron transfer between *N,N*-dimethylformamide and Li metal generates free radicals that induce the decomposition of the polymer and Li salt, resulting in the formation of a LiF-rich mosaic interface layer. By tuning the growth rate and morphology of this interface layer, the SSBs possess excellent long-term cycling stability.^[164,165,214] As for inorganic SEs, fluorine doping in the SEs has been used.^[184,213] A LiF- and YF_x -containing buffer layer with a homogenous morphology was in situ formed between a fluorine-doped bromide SE and Li metal during cycling, and it successfully inhibited the growth of Li dendrites. Such an in situ fluorinated layer was superior to an artificially fluorinated layer on the surface of Li metal in terms of the homogeneity

of fluoride distribution and intimate contact with the Li-metal anode. The improved interfacial stability led to the increase of the critical current density and high cycling stability of the $\text{Li}[\text{Li}_3\text{YBr}_{5.7}\text{F}_{0.3}]\text{LiCoO}_2$ cells at RT.^[184]

Second, modifying the structure of the anodes is useful in suppressing the growth of Li dendrites. Typical Li anodes have a planar morphology. As a result, the contact area with the SE is limited. Besides, the growth of Li dendrites from such a planar anode is inevitably perpendicular to the Li/SE interface, easily causing the penetration of SEs.^[215] In comparison, rationally designed morphologies such as the 3D anode and vertically aligned anode not only can significantly increase the contact area with the SEs, but also may alter the direction of Li dendrite growth (Figure 5f).^[215–217] A vertically aligned Li anode with lithophilic microwalls was able to induce the lateral growth of deposited Li, thus preventing the Li dendrite penetration through the poly(ethylene glycol) dimethyl ether-matrix SE separator.^[215] A 3D Li-Ni composite anode possessed the increased number of Li deposition sites and decreased effective current density with respect to the Li-metal anode. A uniform Li nucleation without Li dendrite growth was thus successfully achieved.^[217]

At last, alternative anode materials may be used. Besides pure Li metal, some Li-containing alloys such as Li-In, Li-Al, Li-Zn, Li-Mg, Li-Si, and Li-Sn alloys have been used as anodes in SSBs as well.^[99,218–222] These alloy anodes have several advantages, such as higher redox potentials to avoid the decomposition of SEs (Figure 5g), higher Li-ion diffusion coefficients, and better wetting on SEs.^[219,223,224] For example, a $\text{Li}_{0.8}\text{Al}$ alloy anode showed excellent compatibility toward the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ SE separator and can prevent the reductive decomposition of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$.^[222] Besides the Li-containing anodes, the Si-based anode is a promising alternative in the SSBs due to its ability to form a stable and passivating SEI.^[225] Various kinds of Si anodes have been developed and used in the SSBs, including micro-sized Si, nanostructured Si, and thin-film Si.^[113,225–228] It should be noticed that the Si anode has a tremendous volume expansion up to $\approx 300\text{--}400\%$ at a fully lithiated state, causing severe stability issues such as Si pulverization and electrode cracking.^[229]

5. Interfaces between Current Collectors and Electrodes

The interfaces between the electrodes and current collectors (i.e., copper (Cu) and aluminum (Al)) in SSBs also cause issues that do not exist in commercial LIBs based on liquid electrolytes. When a composite cathode made of sulfide electrolytes, CAMPs, and electronic conductive agents is in direct contact with a smooth, flat Al foil, gaps may form between the composite cathode and the Al current collector due to the rough surface of the solid-state composite cathode. Such a poor contact would lead to decrease of the contact area, increase in the interfacial resistance, and degradation of the cell performance.^[230] To improve the contact area and adhesion of the current collector/composite cathode interface, carbon materials can be used as the current collector. A current collector made of flexible carbon nanotubes (CNTs) with excellent electrical properties showed a much stronger adhesion to the composite cathode compared with a conventional Al collector.^[230] The gaps between the particles in the composite cathode are filled by the

CNTs, leading to larger contact area between the composite cathode and the current collector. Due to the intimate contact, the interfacial resistance between the CNT current collector and the cathode is greatly reduced, consequently enhancing the rate capability and reversible capacity of the SSBs. Another example is a 2 μm thick CNT thin layer coated on the TiS_2 -based composite cathode by a solution-coating process. As a conformal current collector, this CNT layer ensured good electronic contact with the cathode.^[231]

For the composite cathodes containing halide SEs such as Li_3InCl_6 , it has been found that the halide electrolyte can react with Al metal, leading to the decomposition of Li_3InCl_6 and the corrosion of the Al current collector.^[232] To prevent the direct contact between Li_3InCl_6 and Al, a carbon-based coating can also be applied on the Al current collector, which prevents the side reactions at the Al/cathode interface. As a result, the electrochemical performance of the Li_3InCl_6 -based SSB, including the rate capability, reversible capacity, voltage polarization, and cycling stability, was notably improved.^[231]

As for the composite electrodes containing polymer-matrix SEs, it is also necessary to achieve a tight binding between the electrodes and the current collectors. A hybrid SE containing poly(vinyl alcohol)-g-pyrrole-2-carboxylic acid, LiTFSI, ionic liquid, and pyrrole-2-carboxylic acid modified hydrotalcite nano-conductor was synthesized and utilized as a conductive binder in a LiCoO_2 cathode.^[233] The strong hydrogen bonding originating from the huge number of hydroxyl groups ($-\text{OH}$) in the hybrid SE generates a large adhesive strength ($>24\text{ MPa}$) between the cathode and the Al foil. The adhesion between the cathode and the Al current collector is robust and allows for rapid electron transfer at the interface. In addition, an integrated structure of the electrode and current collector is also helpful for reducing the interfacial resistance. A unique gradient composite cathode was prepared.^[234] One side of the composite cathode contained more electronically conductive nanowires and reduced graphene oxide, enabling rapid electron transport and acting as the current collector. The design of the integrated interface between the cathode and the current collector provided a large interfacial contact area and a continuous channel for rapid electron transport.

To ensure robust interfacial binding between the Li anode and the Cu current collector, the roll-to-roll method that is compatible with large-scale industrial production is the preferred approach. However, applying this technique to the Li foil is especially challenging, as Li metal is very adhesive. To laminate the thin Li foil well onto a Cu current collector, an auxiliary membrane (e.g., a polyethylene terephthalate film coated with silicone oil) that exhibits a weak adhesive strength to the Li foil and thus can be easily removed after attachment could be used to facilitate the processing.^[12] Directly evaporating or sputtering metallic Li onto the Cu foil may also be a promising strategy to obtain a Li-metal anode with the desired thickness and quality.^[235]

6. Outlook

SSBs are continuing to attract ever-increasing attention owing to their promising high energy density, long lifespan, and high safety. The core of SSBs is the solid-state cell that

is a multilayered (e.g., Cu|anode|SE separator|composite cathode|Al), multiphase composite system with numerous solid–solid interfaces at multiple scales. The performance of SSBs is largely controlled by these solid–solid interfaces, including internal buried interfaces within SEs and composite electrodes, and planar interfaces between electrodes and SE separators or current collectors. These solid–solid interfaces may suffer from slow ion/electron transport and poor chemical/electrochemical compatibility, while the solid–solid contact is also vulnerable to the volume change of active materials during cycling. Besides, when Li metal is used as the anode, Li dendrites may emerge after repeated cycling and compromise the interface as well.

Remarkable achievements have been made in addressing the interfacial issues in SSBs, but many challenges still exist. To obtain stable and benign interfaces in SSBs for long-term cycling, novel materials, assembly processing, interface engineering methods, and characterization techniques need to be developed, while the effective accomplishment of these goals demands in-depth mechanistic studies. Our perspective on the future investigation and optimization is described below (Figure 6).

1) *Conductive, dynamically stable interfaces*: Since the fast charge carrier transport is required for each interface in SSBs, it is critical to construct compact, stable solid–solid contact that exhibits low interfacial resistance and excellent tolerance to long-term cycling. To achieve this goal, SEs with optimal electrical, mechanical, and (electro)chemical properties should be designed, and the manufacturing of solid-state cells should also be taken into account. At present, most proof-of-concept solid-state cells reported in the literature use a low mass loading of active materials in the composite cathodes and an overly thick layer for the SE separators. Although such cells can very well serve the purpose of basic research, their energy densities are not competitive. When the mass loading of active materials is increased to meet the energy-density requirement for practical applications, the interface

issues in the composite cathodes, especially those related to long-term cycling, would become more prominent. This is specially so when nanosized active materials are used. Therefore, sufficient attention must be paid to the maintenance of the dynamic stability of the interfacial contact upon long-term cycling of solid-state cells.^[236–238]

- 2) *In situ and operando characterization techniques*: During the operation of SSBs, the solid–solid interfaces undergo not only charge transfer, but also electrochemical reactions. Developing advanced in situ/operando characterization techniques to monitor the evolution of the composition and structure of the interfaces is helpful to deeply understand the mechanisms of both interfacial behaviors mentioned above, while such knowledge would eventually serve as important guiding principles for the design and fabrication of SSBs. Some in situ/operando technologies have been employed to monitor the evolution of the interfaces in real time and to elucidate the mechanisms of the interfacial reactions in SSBs, including operando STEM with electron energy-loss spectroscopy, in situ and operando X-ray computed tomography, in situ neutron depth profiling, and in situ magnetic resonance imaging.^[15,171,239–248] Such nondestructive techniques provide critical support in studying the solid–solid interfaces. Nevertheless, at present only a limited number of materials have been investigated by these advanced characterization techniques. In order to reach a comprehensive understanding on the interfacial behaviors, such in-depth study needs to be conducted on as many materials systems as possible.
- 3) *Theoretical calculations*: Theoretical calculations such as density functional theory-based calculations, ab initio molecular dynamics, finite elemental modeling, and simulations based on various battery models have been used to predict the reactions and evolution of the solid–solid interfaces, to analyze the Li plating/stripping behaviors in SSBs, to study the space charge layer effects, and to search for proper coating materials.^[88,249–256] With the assistance of theoretical calculations, tremendous trial-and-error-based efforts can be saved. To

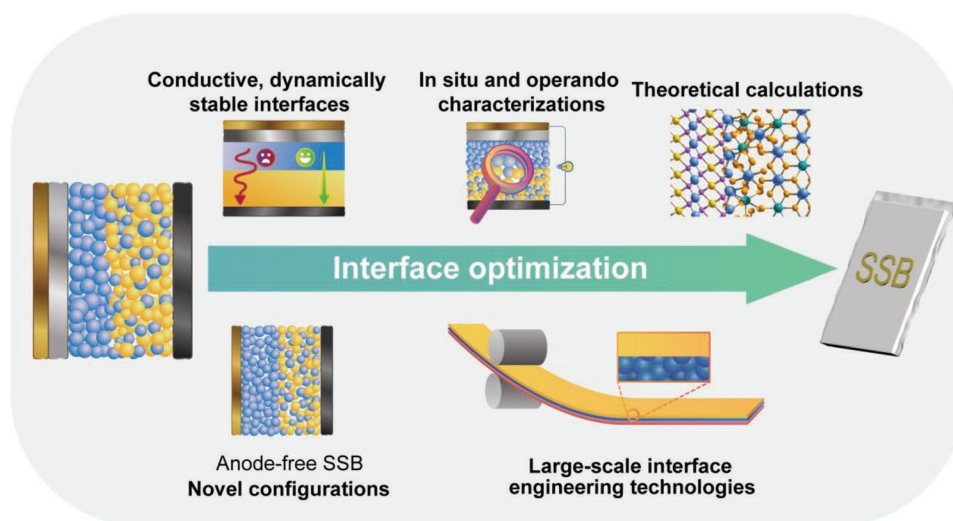


Figure 6. Perspectives for the future research on the interfaces of SSBs.

quickly narrow the scope of candidate materials with desirable properties, high-throughput computing and the establishment of large computational materials databases are necessary.^[88,255] It is also important to develop the multiscale simulation models that take into account various kinetic processes and the calculation methods that are well integrated with experimental results.^[88,255] The continuous progresses in theoretical calculations will significantly promote the design and optimization of the interfaces in SSBs.

- 4) **Novel battery configurations:** The single solid-state cells with various configurations have been designed and fabricated in the literature, demonstrating remarkable electrochemical performance. Anode-free solid-state cells without the highly reactive Li-metal anode have the advantages of high energy density and safety, thus attracting a lot of interest.^[257–260] The performance of the anode-free SSBs is dependent on the composite electrode/SE interface and the SE/current collector interface; the quality of these interfaces is closely related to the morphology of the current collector, the buffer layer, the properties of SEs, and even the applied pressure.^[259–262] At present, anode-free solid-state cells still suffer from low Coulombic efficiency and poor reversibility, which result in a short cycle life at RT. This issue shall be resolved through materials development and interface engineering. Besides the anode-free configuration, a monolithic configuration with sulfides as both the active materials and the electrolytes can increase the number of redox active sites for reversible electrochemical reaction and avoid undesired interfacial reaction between the cathodes and the SEs.^[263–265] New battery configurations may be designed to significantly enhance the energy density and safety of the SSBs and to reduce the cost of materials and production.
- 5) **Large-scale interface engineering technologies:** The SSBs are still immature for mass production due to their complicated and expensive production processes. Unlike the large-scale production of conventional LIBs, one single technology process chain that is universally applicable to the mass production of all kinds of SSBs is not likely to exist. Instead, different approaches might be in operation for the production of SSB cells, depending on the selection of SEs.^[12,266] No matter which approach is adopted, it is necessary to fabricate high-quality interfaces, and the achievement of this goal usually demands interface modification. Therefore, developing facile, low-cost, and large-scale interface engineering technologies is very important for the manufacturing of practical SSBs. The possibility for mass production shall be considered in the research and development of interface modification for the SSBs.

The role of the interfaces in SSBs is manifold and critical. A considerable number of efforts are still needed for the study of this subject. If dynamically stable, benign interfaces can be realized in the entire solid-state cell with reasonable cost, the extensive applications of SSBs would be highly expectable.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

composite cathodes, grain boundaries, interfaces, lithium anodes, solid electrolytes, solid-state batteries

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